Mind Scope: Prediction Of Mental Health Using Machine Learning Techniques

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ABSTRACT

- Mental health issues such as anxiety, depression, and stress have become increasingly prevalent due to lifestyle changes, academic pressure, and social factors. Early detection and intervention can significantly improve treatment outcomes and overall well-being. This project presents a Mental Health Prediction System using the Naive Bayes algorithm, a probabilistic machine learning model known for its simplicity and effectiveness in classification problems. The system analyzes user data including demographic information, lifestyle habits, work or academic stress levels, and self-reported symptoms to predict the likelihood of a mental health condition. The Naive Bayes classifier calculates the probability of different mental health states based on prior occurrences, enabling accurate and fast predictions even with limited datasets. The model's performance is evaluated using metrics such as accuracy, precision, recall, and F1-score, demonstrating reliable results. This predictive system can assist psychologists, counselors, and health organizations in identifying at-risk individuals early and providing timely mental health support.

INTRODUCTION

- Mental health issues such as anxiety, depression, and stress are increasing due to lifestyle changes, academic pressure, and social influences.
- Early detection and intervention are essential to improve treatment outcomes and overall well-being.
- The project uses the Naive Bayes algorithm, a probabilistic and efficient machine learning model, to predict mental health conditions.
- User data including demographics, lifestyle habits, stress levels, and self-reported symptoms are analyzed to estimate the likelihood of mental health issues.
- The system's performance is measured using accuracy, precision, recall, and F1-score, helping psychologists and health professionals identify at-risk individuals early.

PROBLEM STATEMENT

- Mental health issues such as anxiety, depression, and stress are increasing due to lifestyle changes, work pressure, and social factors.
- Traditional diagnosis methods are often time-consuming, subjective, and lack accessibility for early detection.
- There is a growing need for an automated, data-driven system to identify individuals at risk of mental health problems.
- Machine learning techniques can help analyze user data to predict mental health conditions efficiently and accurately.
- This project proposes a Mental Health Prediction System using the Naive Bayes algorithm to enable early detection and timely intervention for better mental well-being.

EXISTING SYSTEM

- Existing systems for mental health prediction mainly rely on survey-based assessments and traditional machine learning models that use limited features such as smartphone usage or self-reported data. These systems often face major challenges like small or short-term datasets, lack of interpretability, and population-level analysis that fails to capture individual differences. For example, the I-HOPE (Interpretable Hierarchical mOdel for Personalized mEntal Health Prediction) developed by Roy Chowdhury et al. (2025) addresses some of these limitations by using behavioral data such as sleep patterns, phone usage, and social interactions from the College Experience Study (CES) dataset to provide personalized insights. While earlier models achieved only 60–70% accuracy, I-HOPE reached 91% accuracy by applying hierarchical modeling and interpretable predictions. However, these systems still face limitations such as dependency on continuous data collection, missing data issues, and computational challenges for real-time implementation.

DISADVANTAGES

- Dependence on Self-Reported Data
- Limited Dataset Size and Duration
- Lack of Model Interpretability
- Absence of Personalization
- Data Privacy and Computational Challenges

PROPOSED SYSTEM

- The proposed system aims to develop a Mental Health Prediction Model using the Naive Bayes algorithm to identify individuals at risk of mental health issues such as anxiety, depression, and stress. The system analyzes various user attributes, including demographic details, lifestyle habits, work or academic stress levels, and self-reported symptoms, to estimate the likelihood of developing a mental health condition. The Naive Bayes classifier, known for its simplicity and high efficiency, calculates the probability of mental health states based on prior occurrences, enabling accurate predictions even with limited datasets. The model's performance will be evaluated using key metrics such as accuracy, precision, recall, and F1-score to ensure reliability. This predictive system is designed to assist psychologists, counselors, and healthcare professionals in detecting at-risk individuals early, providing timely intervention, and improving overall mental well-being.

ADVANTAGES

- ✈Early Detection
- ✈High Accuracy
- ✈Fast and Efficient Processing
- ✈Simplicity in Implementation
- ✈Enhanced Decision Support

SYSTEM REQUIREMENTS

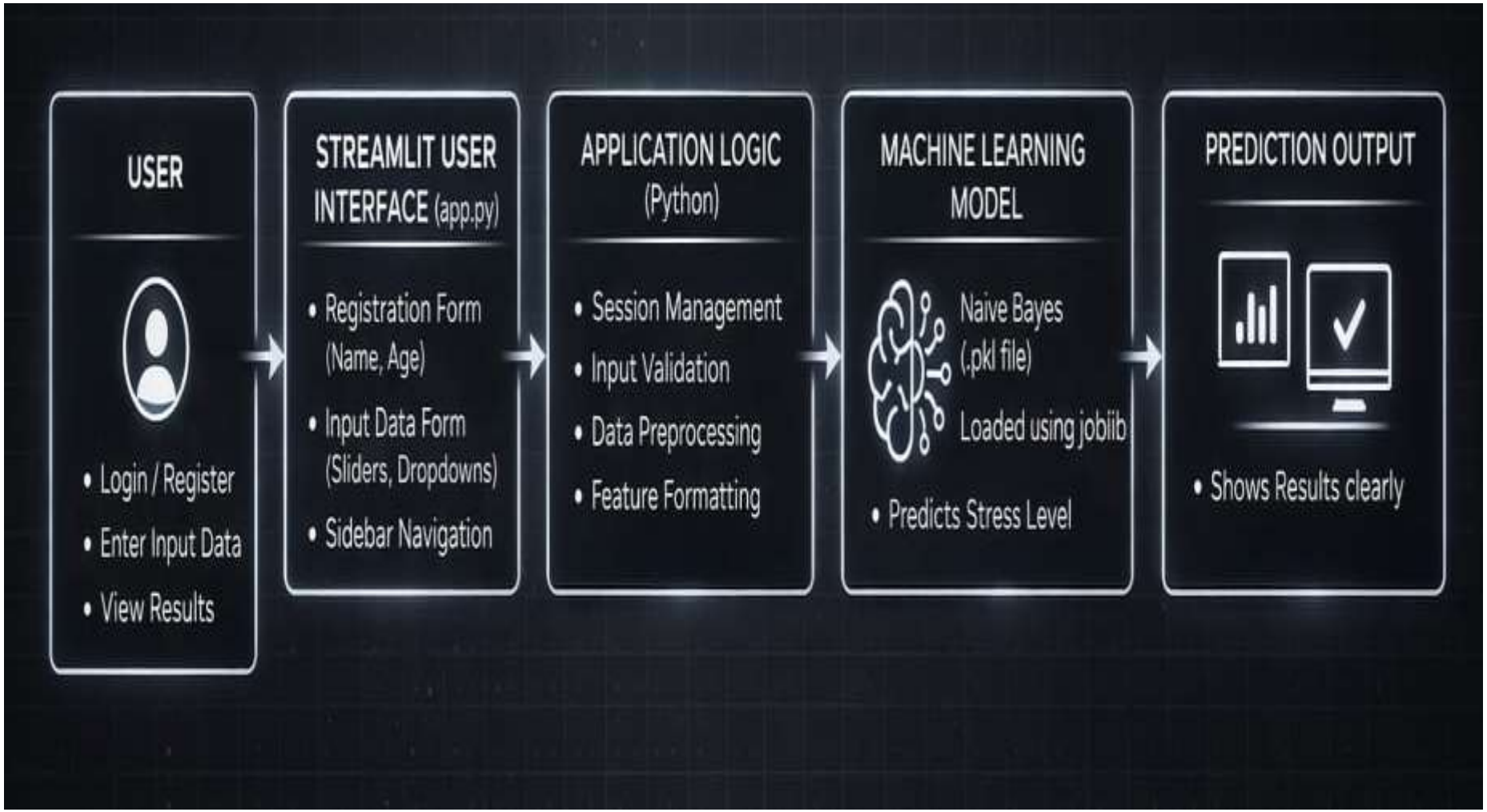
Hardware Requirements:

Processor	: Intel i3 / Ryzen 3 or above
RAM	: 4 GB or more (8 GB recommended)
Storage	: 20 GB free space
Display	: LCD/LED Monitor
Input Devices	: Keyboard and Mouse

SOFTWARE REQUIREMENTS

Operating System	: Windows 10 / 11
Programming Language	: Python
Framework	: Streamlit
Frontend	: Streamlit UI
Libraries	: Scikit-learn, Pandas, NumPy, Joblib
IDE	: VS Code

System architecture



MODULES

- User Registration module
- Data input module
- Data preprocessing module
- Prediction module
- Result display module
- Recommendation module

MODULES

User Registration module

- Collects basic user details such as name and age
- Stores user information using session state
- Provides personalized interaction within the application
- Enables smooth navigation across different sections

MODULES

Data Input Module

- Allows users to enter mental health-related data
- Includes parameters like stress level, sleep quality, anxiety, etc.
- Uses sliders and dropdowns for easy input
- Ensures user-friendly and interactive interface

MODULES

Data preprocessing module

- Converts user input into structured numerical format
- Handles data cleaning and formatting
- Ensures compatibility with machine learning model
- Prepares features required for prediction

MODULES

Prediction module

- Uses trained Naive Bayes model for classification
- Processes input data to predict stress level
- Provides fast and efficient predictions
- Suitable for small datasets and real-time usage

MODULES

Result display module

- Displays prediction results to the user
- Shows stress level (Low, Medium, High)
- Provides clear and understandable output
- Enhances user experience with visual feedback

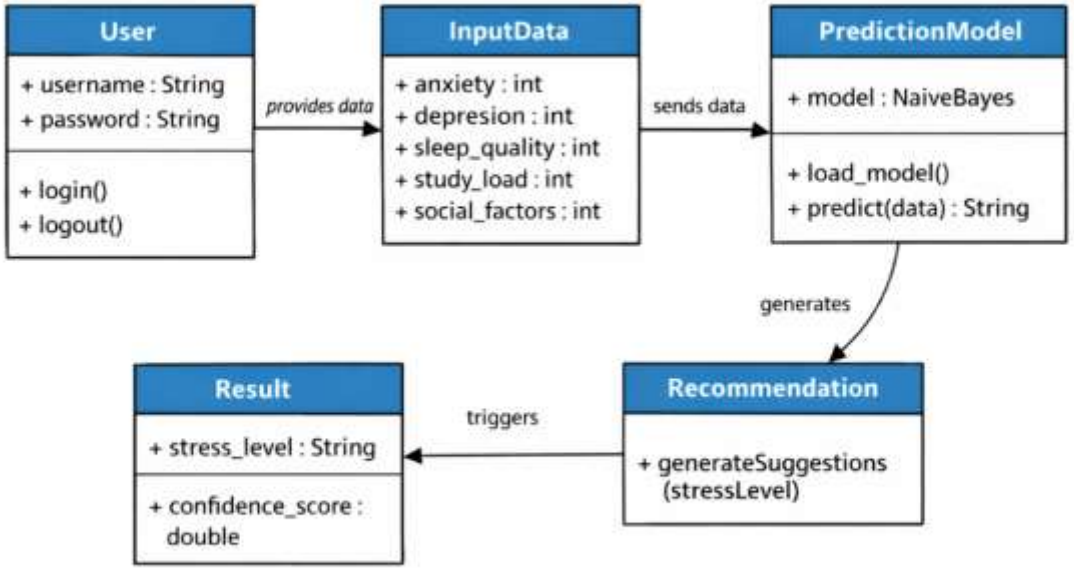
MODULES

Recommendation module

- Provides personalized wellness suggestions
- Suggests activities like exercise, meditation, and relaxation
- Helps users improve mental well-being
- Supports early intervention and awareness

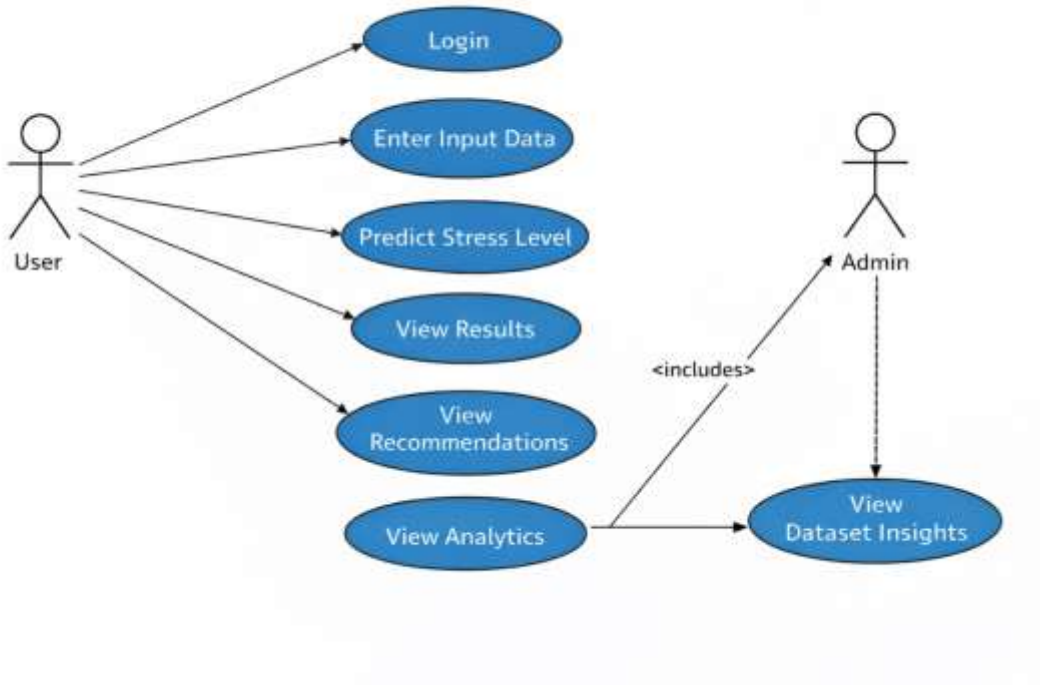
UML Diagrams

class diagram



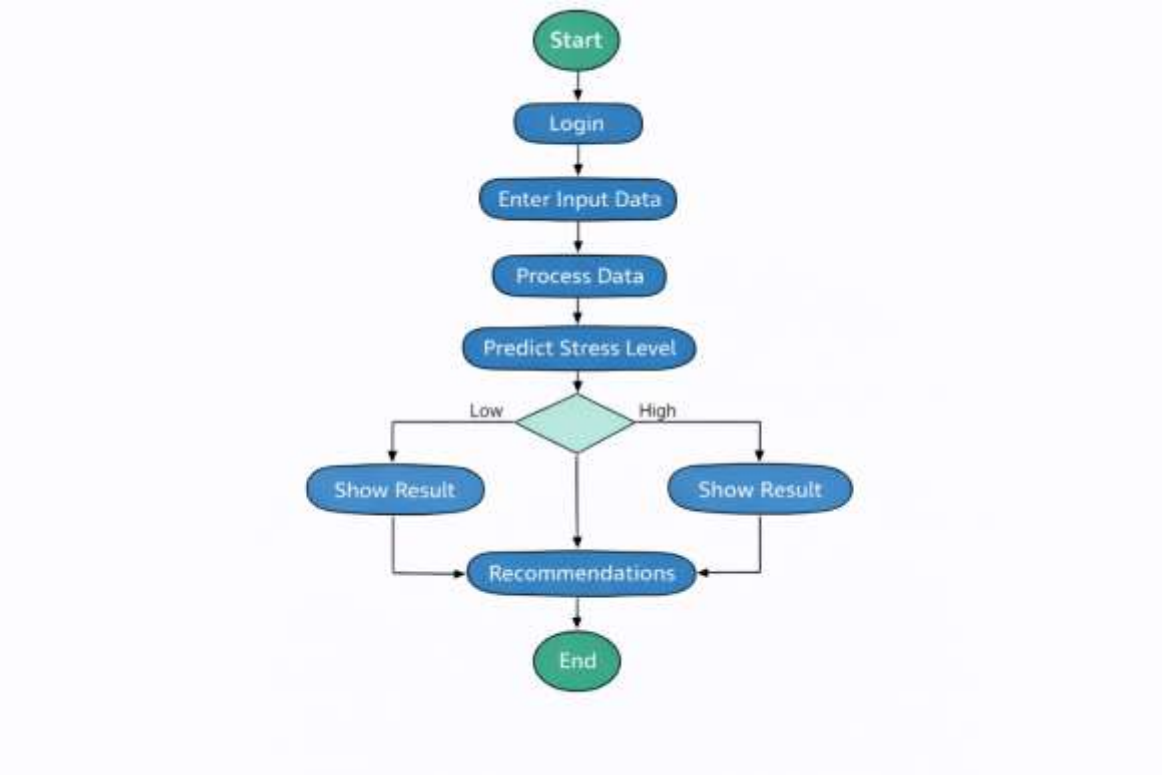
UML Diagrams

Usecase diagram



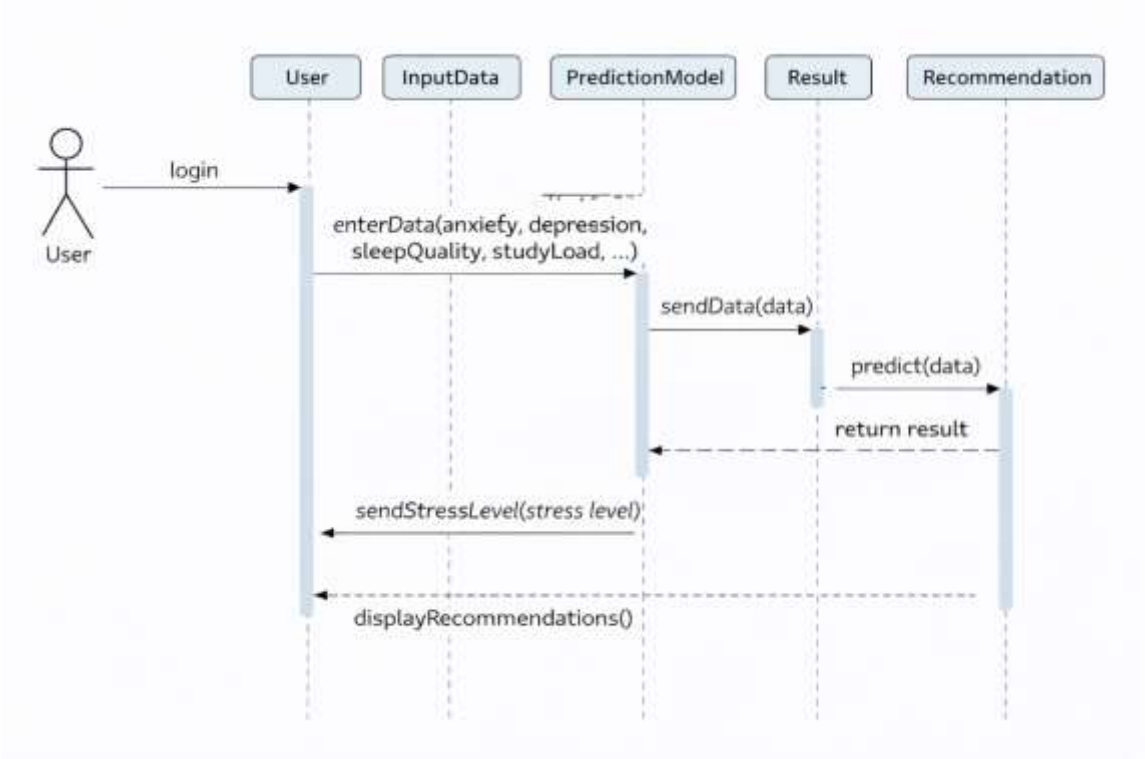
UML Diagrams

Activity diagram



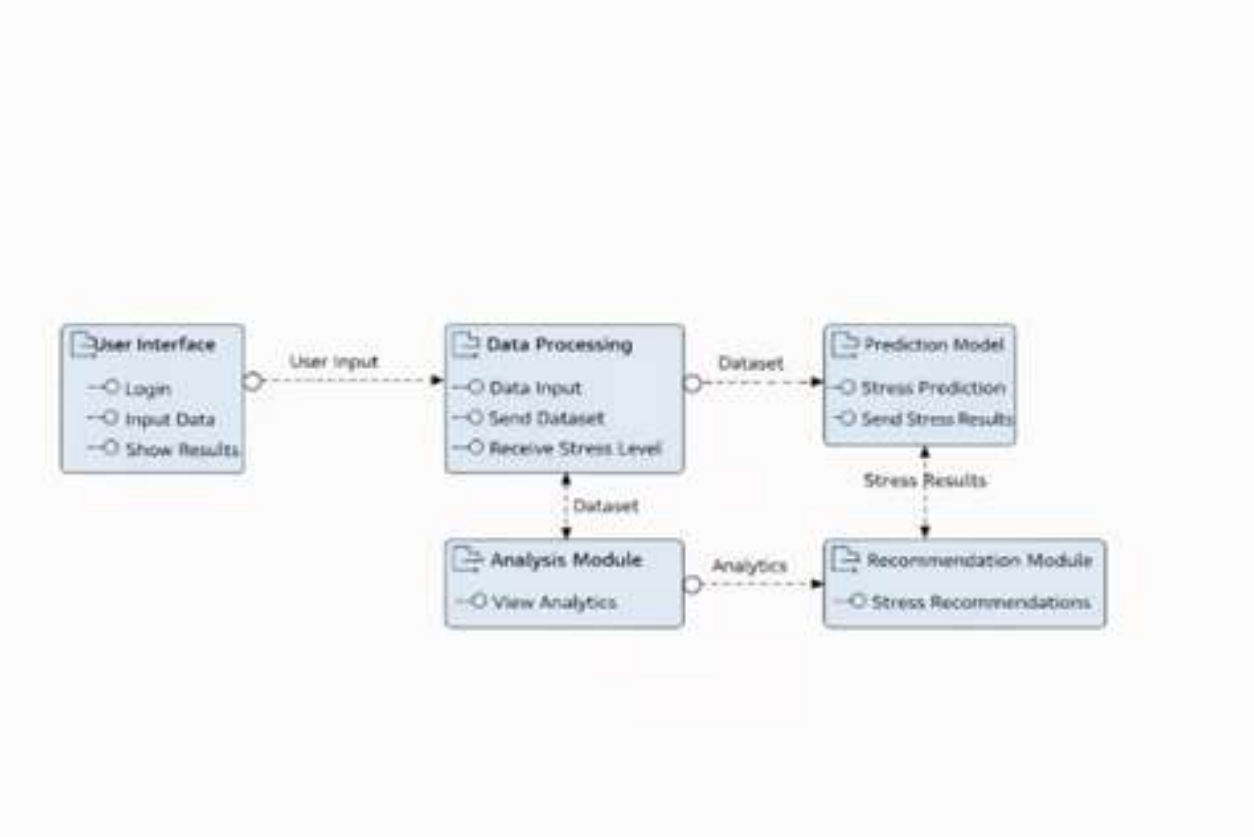
UML Diagrams

Sequence Diagram



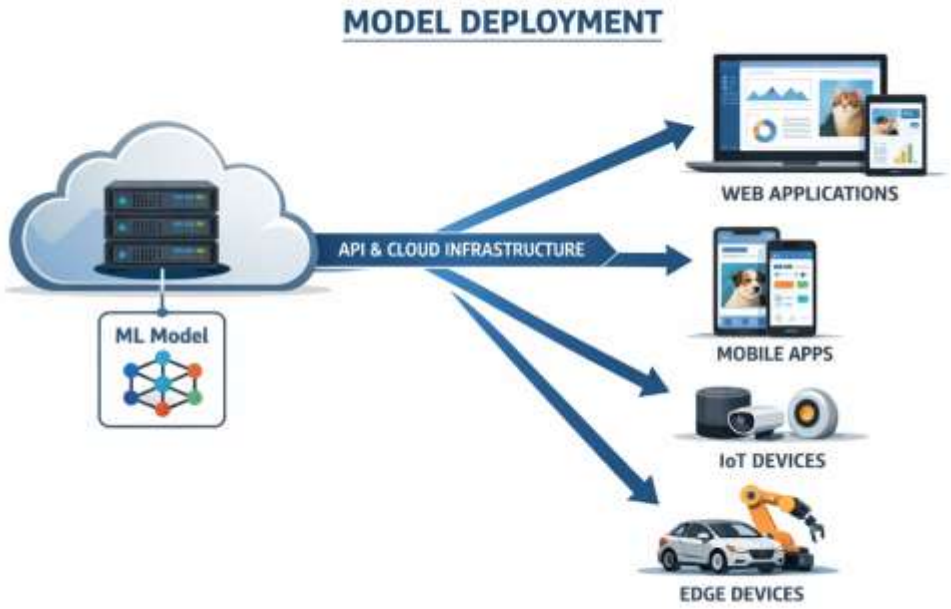
UML Diagrams

Component Diagram



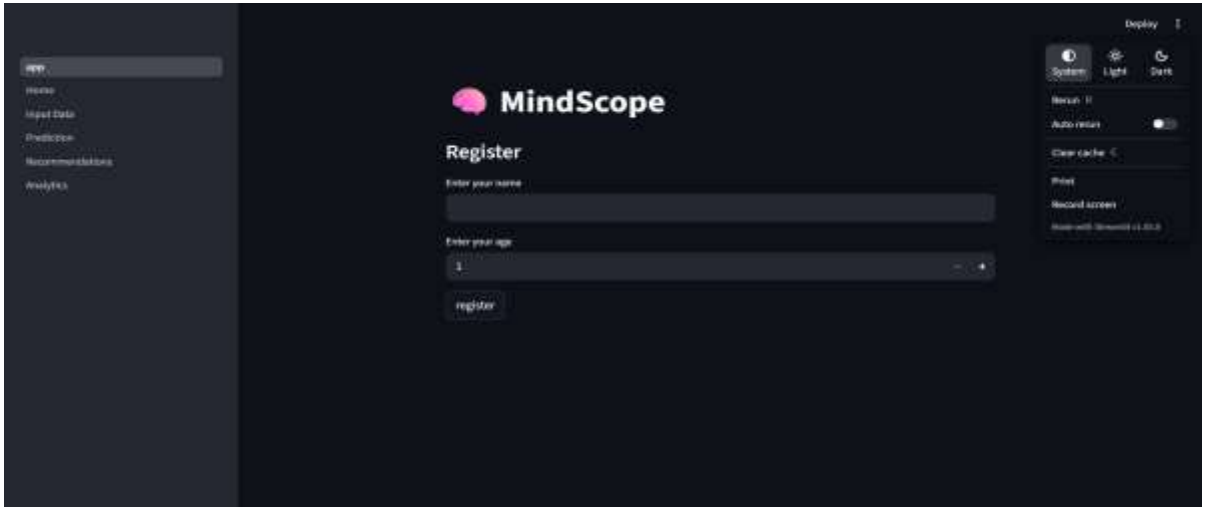
UML Diagrams

Deployement Diagram



Output

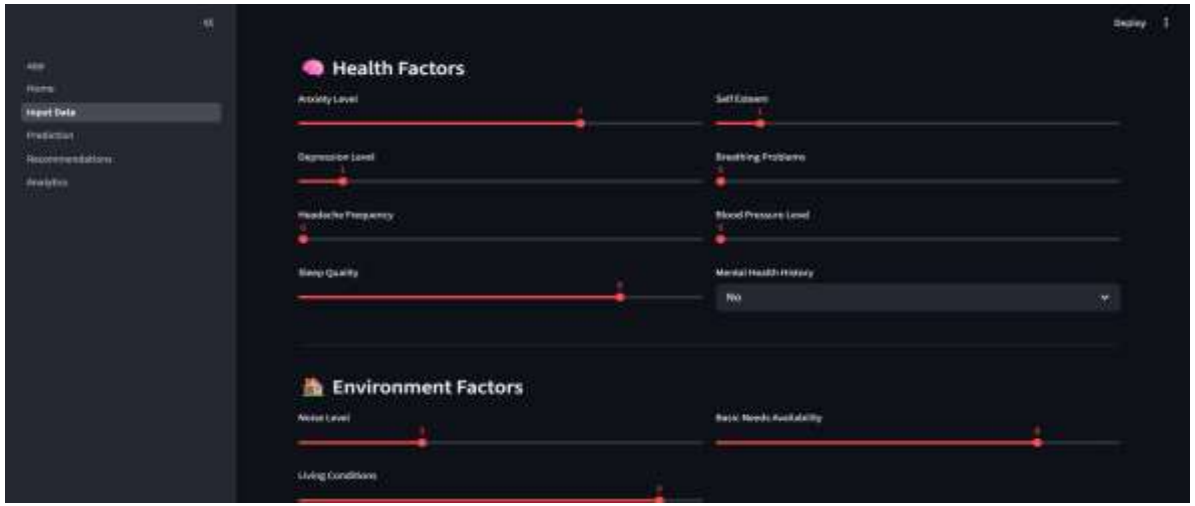
Interface





Output

Input

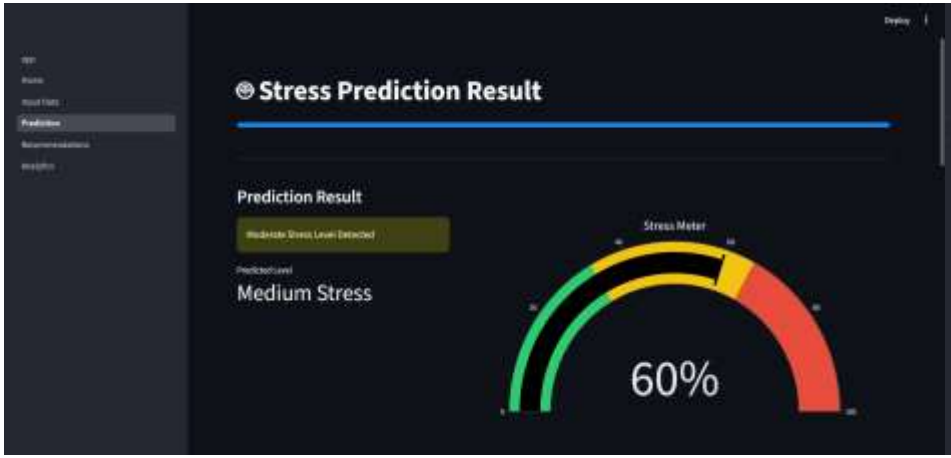




Output

prediction

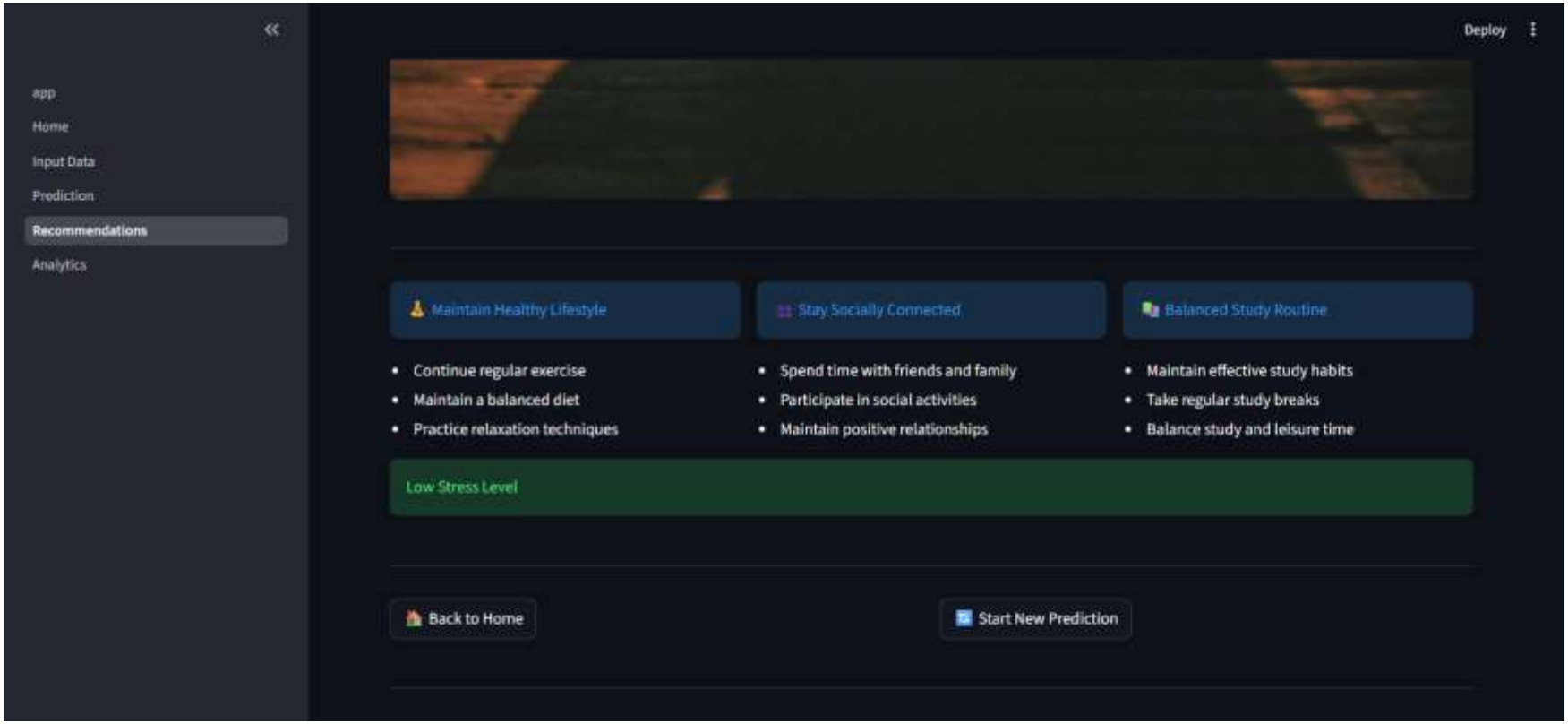






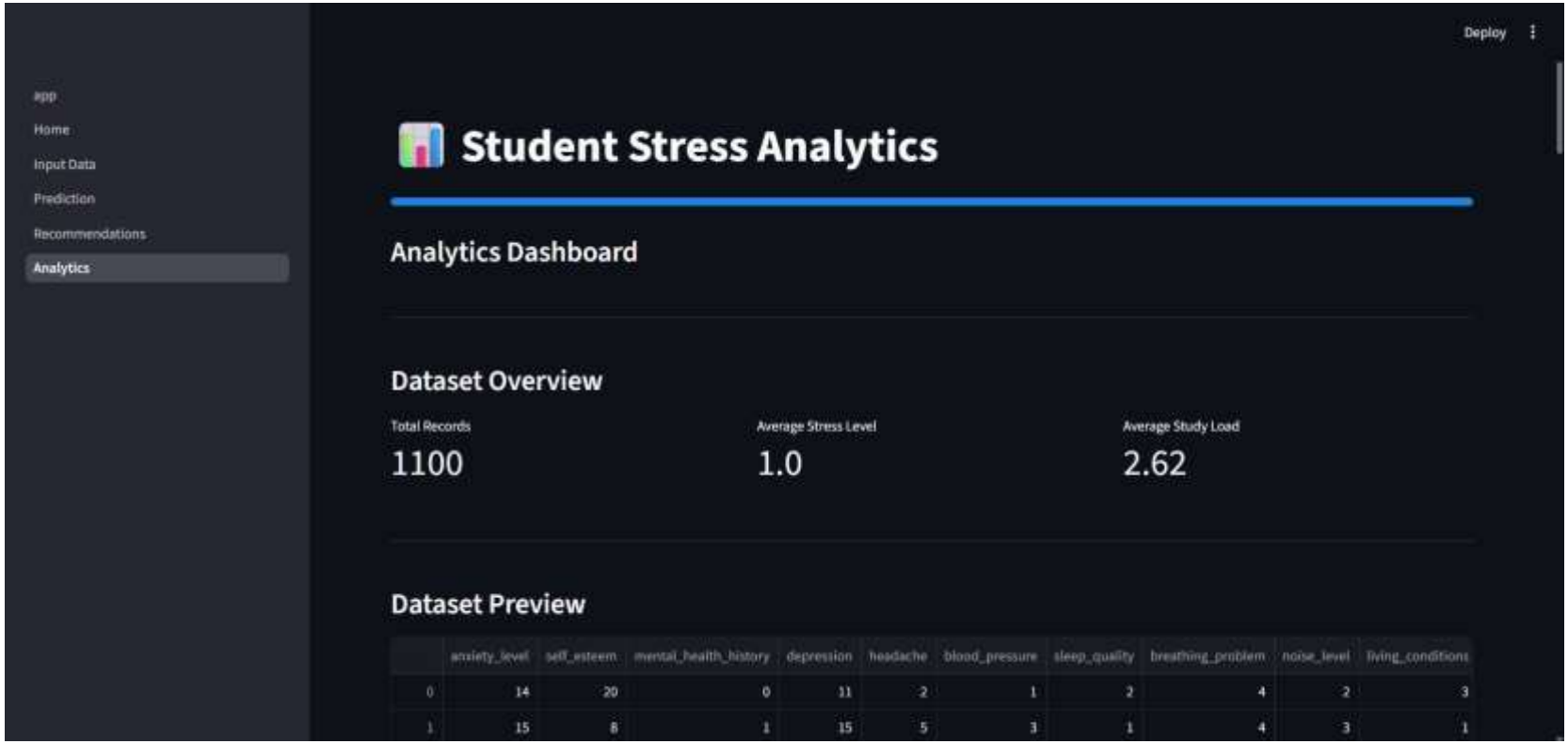
Output

Recommendations



Output

analytics



Future Enhancements

The current system provides an effective solution for mental health prediction. However, there is scope for further improvements to enhance accuracy, usability, and functionality. The following enhancements can be implemented in future to make the system more advanced and efficient.

Model Improvements

- Use advanced algorithms like Random Forest and Deep Learning
- Improve prediction accuracy and performance
- Train model with larger and diverse datasets

System Features

- Add real-time data tracking and monitoring
- Enhance user interface for better experience

Future Enhancements

Security & Data

- Implement user authentication (login system)
- Add secure data storage and privacy protection

Platform Expansion

- Develop mobile application version
- Enable cloud deployment for wider accessibility

Advanced Integrations

- Integrate chatbot for mental health support
- Connect with wearable devices (fitness trackers)

CONCLUSION

• The proposed system successfully predicts mental health conditions using the Naive Bayes algorithm.

• It provides a simple and user-friendly interface through Streamlit for easy interaction.

• The model delivers accurate and efficient predictions based on user input data.

• The system helps in early identification of stress levels, supporting better mental health awareness.

• Overall, the project demonstrates the effective use of machine learning in real-world healthcare applications.

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Thank you!